

A Voxel-based Grid Coordinate for Virtual Object Positioning in Outdoor Mobile Augmented Reality

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ABSTRACT

Humans perceive and describe space using relative expressions, while computers use absolute coordinates. This difference creates a gap in spatial communication, especially in outdoor environments where objects with complex surfaces serve as references. To address this, we present a voxel-based surface grid coordinate system for outdoor mobile augmented reality. This approach voxelizes reference object models obtained from a 3D map and defines grids on them. Each surface grid maps to geographic coordinates and provides a surface-based index for placement. Quantitative outdoor experiments and a pilot user study show how surface grids and visual cues support spatial perception at different levels.

Index Terms: Augmented reality, Human-computer interaction, User interfaces

1 INTRODUCTION

Humans perceive and describe space in relative terms (e.g., in front of an entrance gate), whereas computational systems represent space using absolute coordinate systems such as latitude, longitude, and altitude. This discrepancy creates a fundamental gap in spatial communication between humans and computers. Augmented Reality (AR) can be a solution to this by creating an intermediate representation that is understandable to both.

A basic method for communicating spatial information in AR is the placement of virtual objects, and it needs reference object surfaces to anchor them. Especially in outdoor AR, surface scanning is unreliable due to sensing limitations at large distances. As a result, recent outdoor AR systems commonly rely on pre-built 3D models and geospatial anchoring to place virtual content at city scale through map-based localization pipelines [1, 4, 6].

While these model-based approaches primarily focus on positioning accuracy, reference objects with complex or curved surfaces are used as they are, without providing stable surface-level coordinates that align with human spatial reasoning. This makes it difficult to express and interpret relative spatial cues such as local position, scale, and distance with respect to reference object surfaces.

Previous work has shown that visual aids such as grids can improve distance estimation and alignment accuracy [2, 5]. However, in outdoor AR, such grids cannot be directly defined on complex, curved reference object surfaces.

Therefore, we introduce a voxel-based surface grid coordinate system for outdoor mobile AR. Using voxelization, we approximate complex surfaces as locally planar regions and define grids on them for expressing relative positions, distances, and directions, while maintaining a direct correspondence with geographic coordinates. We also provide guide lines that support surface attachment and depth perception during placement. We conduct a quantitative outdoor placement experiment for evaluation and a pilot user study to get insights for further development.

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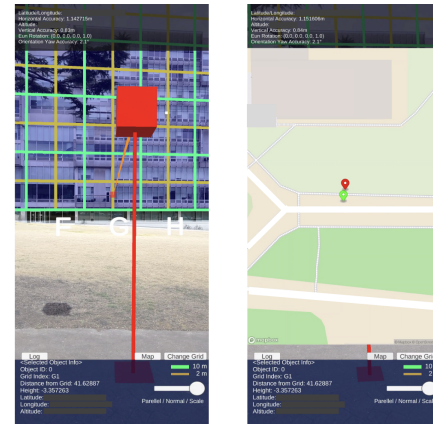


Figure 1: (Left) main screen, (right) map screen

2 METHOD

Voxel-based Surface Approximation Real-world reference objects have complex and curved surfaces, which makes it difficult to define grids directly on them. We address this by applying voxelization to the reference object model.

In detail, we divide the object into small cubic voxels and use the exposed voxel faces as surface elements. Note that the unit length of the voxels is chosen to keep voxelization error under a predefined threshold. Each face provides a small flat region on which a grid can be defined. By aggregating these faces, complex outdoor object surfaces can be handled as a set of simple surface patches suitable for grid-based spatial reference.

Surface Grid Coordinate Construction and Anchoring We assume that the reference object model is obtained from a 3D map such as OpenStreetMap. Based on the voxelized surfaces, we build a grid directly on the reference object surface. Each grid cell provides a surface-based index for referring to locations on the object (e.g., A1-1). The surface grid is anchored to geographic space by mapping each grid coordinate to a specific latitude, longitude, and altitude, allowing surface-based references to be directly translated into geographic coordinates.

To support different levels of spatial detail, the grid can be divided further to provide finer resolution without additional voxelization. As a result, the same surface can support multi-scale spatial reference.

Interface Design for Spatial Perception To support the use of the surface grid in mobile AR, we design interface elements that aid spatial perception and placement.

Normal lines are drawn from a virtual object to the surface grid and the ground to indicate surface location and depth. The "shadow" is vertically projected onto the surface grid to show its size and occupied area. A 2D map view shows the relative positions of the user and virtual objects in geographic space.

For object manipulation, movement is constrained to directions

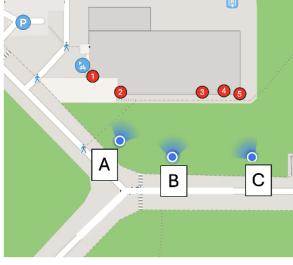


Figure 2: Test locations

	Mean	Std	Min	Max
Horizontal Placement Error (m)	6.27	2.19	2.43	9.74
Vertical Placement Error (m)	1.95	1.32	0.23	4.53
Normalized Horizontal Placement Error (%)	6.96	2.43	2.70	10.82
Normalized Vertical Placement Error (%)	4.87	3.30	0.57	11.32

Table 1: Summary

parallel and perpendicular to the surface grid. This makes placement follow the surface grid rather than the user's viewpoint.

3 QUANTITATIVE EVALUATION

Experiment We conducted a placement accuracy evaluation to verify the proposed method. The system was implemented as a mobile AR application running on a commercial smartphone (Samsung Galaxy S22) using Unity AR Foundation with ARCore. A reference object model was obtained from OpenStreetMap and voxelized into 10 m units. Based on this representation, a surface grid with 2 m and 5 m sub-grid resolutions was built and anchored to geographic coordinates using ARCore's geospatial API.

For evaluation, five target points were set on different surfaces of the reference object building at different height levels. The ground truth locations of these target points were measured using a professional GPS device (E-Survey Esa2) with 1408 channels that provides horizontal accuracy within 0.50 m and vertical accuracy within 0.30 m. Virtual objects were then placed at each target point from three different locations using the proposed interface, as shown in Fig. 2. The geographic coordinates of the placed virtual objects were recorded and compared against the measured ground truth locations to compute placement errors.

Results Tab. 1 summarizes the overall placement accuracy across all experiments in both absolute and normalized terms. The proposed method showed an average horizontal placement error of 6.27 m and an average vertical placement error of 1.95 m. When normalized by the scale of the voxelized reference object, the average horizontal and vertical errors were 6.96% and 4.87%, respectively.

4 PILOT USER STUDY

Study Design We conducted a pilot user study to investigate how different visual cues affect spatial perception and placement. Three participants (P1–P3) participated in a within-subject design. Each participant experienced three visualization conditions: *Full* (mesh + grid + guide line), *Mesh-only* (mesh + guide line), and *Plain* (no mesh, no grid, no guide line).

All participants were asked to place virtual objects at two target locations (2 and 5) under all conditions. To reduce learning effects, each participant experienced the conditions in a different order. After completing the placement tasks for each condition, participants had one-on-one interviews consisting of comparative and open-ended questions to gather qualitative feedback on usability.

Results and Analysis The *Plain* condition was consistently rated as the least usable by all participants, with reports of poor distance and

surface perception and low confidence in placement. Preferences between the *Full* and *Mesh-only* conditions varied among participants; however, the perceived roles of individual visual cues were consistent.

Participants reported that the normal guide line was the most critical cue for precise attachment and fine-level localization on the reference surface. The grid primarily supported scale perception and coarse localization, although some participants noted that it often obscured details of the reference object. Overall, the pilot study suggests that different visual cues support spatial perception at different levels of granularity, from coarse to fine, and that interface design should support spatial understanding without obscuring critical details of the reference object. This observation points to an important direction for further development.

5 DISCUSSION AND CONCLUSION

The quantitative evaluation showed that the proposed voxel-based surface grid coordinate system supports consistent virtual object placement in outdoor environments, although placement accuracy can be further improved. Errors caused by GPS inaccuracies and sensor noise [3] remain as limitations and require further improvement. The pilot user study showed that the grid supports scale perception and coarse localization, but its visibility can interfere with understanding the reference object surface. This result indicates that visual elements such as grid thickness and transparency should be adjusted to support spatial perception without hiding important surface details. Overall, the results show that different visual cues support spatial perception at different levels, and that interface designs should adjust these cues based on user distance and task needs.

In summary, we present a voxel-based surface grid coordinate system that defines surface-based spatial references on complex outdoor objects and connects them to geographic coordinates. The system also uses guide lines to support surface attachment and depth perception during placement. Quantitative outdoor experiments and a pilot user study show its feasibility for outdoor mobile AR interaction scenarios.

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